

Khine Zar Thwe

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PROFESSIONAL SUMMARY

Software engineer with production experience shipping full-stack and mobile applications across Java, C#/.NET, Node.js and React, and building LLM applications with Amazon Bedrock, Claude API, RAG and LangChain. Experienced in REST API design, third-party API integration, transactional database work and serverless deployment on AWS, with hands-on ownership of UAT and release quality for apps used by 200+ field staff.

TECHNICAL SKILLS

Programming Languages: Java, Python, JavaScript, C#, SQL, HTML/CSS

Frameworks & Libraries: React, React Native, Jakarta EE, Spring Boot, Node.js/Express, FastAPI, Hibernate, TensorFlow, OpenCV, MediaPipe, YOLO, CNN models, ASP.NET Core, .NET Core, EF Core

Web Technologies: Vite, TailwindCSS, REST APIs

Cloud & DevOps: AWS (Lambda, S3, Bedrock, API), Docker, Git, Linux, CI/CD

Databases: MySQL, PostgreSQL, Supabase, SQL Server

Mobile & Device Integration: Android Studio, Xcode, iOS/Android, Bluetooth, QR scanners, Face++ API, K10/F2 devices, GPS/EXIF

Testing & QA: UAT coordination, Bug severity classification, Bug report writing, Test case design, Regression testing, Manual testing, Postman (API testing)

AI & LLM: Retrieval-Augmented Generation (RAG), LangChain, Prompt Engineering, Google Gemini API, Claude API, Amazon Bedrock, OpenAI Whisper (speech-to-text), Streamlit

EXPERIENCE

Junior AI Developer – Reachfield IT Solutions

Apr 2025 – Apr 2026 | Singapore

- Developed and deployed YOLO-based crowd-counting and object detection system achieving 85%+ accuracy across 30+ concurrent video streams with real-time intrusion detection.
- Built two production React Native apps (Work Order Management, Form Management) shipped to Google Play and TestFlight, integrating Face++ facial verification and deploying across 200+ active field staff on K10 and F2 ruggedised devices with full UAT coordination.
- Owned end-to-end Quality Assurance (QA) across production releases: conducted comprehensive UAT and bug triage, managed severity classification, wrote detailed bug reports and software documentation, and coordinated resolution across backend and mobile teams, ensuring clear technical writing for cross-functional stakeholders.
- Led Java-based application development by migrating Hibernate 5 → 6 and upgrading JDK 11 → 17 on the SGEMS2 platform, rewriting 50+ DAO/Service classes for Jakarta EE compliance, resolving transaction deadlocks to ensure system integrity, and performing security testing while coordinating UAT for 30+ critical bugs.
- Designed software architecture and implemented a robust REST API layer for SGEMS2, optimising SQL queries on 50 K+ records and training 20+ field staff on deployment and device configuration.
- Demonstrated advanced problem-solving and logical reasoning by conducting a root-cause analysis on CNN inference failures, leading to a 15% improvement in model accuracy through strategic hyperparameter tuning and architectural trade-off evaluations.

Freelance Test Engineer & Java Developer – Freelance

2026-04-01 – Present

- Delivered project-based test engineering for mobile and API applications across 2 startup clients, covering functional, regression, and release-readiness testing with Selenium-based automation.
- Designed and maintained Selenium test suites for mobile and API regression testing, ensuring consistent coverage and faster feedback across client release cycles.
- Built and maintained Spring Boot REST services for client backends, implementing business logic, database integration, and API endpoints to support core product features.
- Collaborated directly with startup stakeholders to gather requirements, define test scenarios, triage defects, and document results for cross-functional clarity.

FEATURED PROJECTS

Portfolio AI Chatbot — AWS Serverless (Bedrock + DynamoDB)

- Built a knowledge-grounded AI chatbot on the portfolio site using a vanilla-JS widget, API Gateway, and Python Lambda, with Amazon Bedrock (Claude Haiku 4.5) generating responses grounded in a system prompt of personal bio, skills, and projects.
- Implemented self-resetting serverless rate limiting in DynamoDB via atomic update_item with ConditionExpression keyed by year-month, capping usage at 300 messages/month with zero cron jobs or scheduled infrastructure.

- Engineered input validation (1000-char limit), conversation-history trimming to the last 6 turns, and graceful quota-exceeded fallback messaging for production-grade UX.

GlucoSG—Full-Stack Nutrition Tracking App

- Full-stack nutrition tracking app for diabetic population built with React/Vite frontend (HTML, CSS, JavaScript), Node/Express backend, and Supabase (PostgreSQL, realtime auth). Developed for a Singapore Polytechnic hackathon and won 1st place.

Code-to-Cloud Architecture Visualizer

- AWS-native application that converts pasted code into architecture diagrams with cost estimates using a React frontend, Python Lambda functions, API Gateway, and Claude Bedrock integration.

EmotionVision—Real-Time Emotion Detection

- Custom TensorFlow 7-class CNN model for emotion classification combined with OpenCV video pipeline and FastAPI microservices, containerised with Docker for real-time emotion and crowd density analysis.

SGEMS2 Enterprise Platform

- Java/Jakarta EE security and estate management system deployed across Singapore. Implemented DAO/Service layer, REST APIs, relational database schema design, JSP templating, and Hibernate ORM. Participated in comprehensive UAT cycle.

Banking QA Automation Framework

- Enterprise-grade end-to-end QA automation framework for banking applications built on ParaBank. Features dynamic test data selection, Page Object Model architecture, and automated testing on every commit via GitHub Actions with live Allure dashboard.

Hearth Coffee Co. — Custom Shopify Theme

2026-08-01

- Built a fully custom Shopify theme on Dawn (Online Store 2.0) with a from-scratch custom section, delivering theme-editor-level customisation in Liquid, JSON, and JavaScript.
- Created a custom “Find your roast” quiz section with no stock equivalent — posts scores to a self-built Node.js/Express API and hydrates results with live Shopify product data, with no Shopify app or Storefront API token required.
- Integrated a self-built service at runtime to demonstrate e-commerce skills beyond a themed storefront, covering custom sections, theme customisation, and API integration.

BookStock — E-Commerce Order & Inventory Management System

- Designed and built a layered (Clean Architecture) REST API in ASP.NET Core handling product catalog, order processing, mocked payments, and stock management, with JWT-based authentication
- Implemented atomic order transactions (stock validation, deduction, and payment creation) using SQL Server transactions with rollback handling to prevent overselling
- Wrote SQL Server stored procedures and views for reporting (monthly sales, top-selling products, low-stock alerts), reducing report query logic outside the application layer
- Deployed to Azure App Service with Azure SQL Database
- Built as a Singapore Polytechnic coursework project, demonstrating full-stack .NET development, database design, and cloud deployment from requirements to delivery.

CERTIFICATIONS

AWS Certified AI Practitioner (AIF-C01) – Amazon Web Services Jul 2026

AWS Cloud Practitioner (CLF-C02) Training Complete – Amazon Web Services Jun 2026

Software Testing/QA – LinkedIn Learning Jun 2026

AI Ethics & Governance (Associate Level) – Singapore Computer Society

S Pass Eligibility Letter – Singapore Polytechnic

EDUCATION

Singapore Polytechnic – Singapore Apr 2026

Diploma in Information Technology (Software & Applications)

Elective: Certificate in Data & Analytics (2024)

LANGUAGES

English (Professional), Burmese (Native)